

Dr. Meena Purohit

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Karnataka- 573201

Date of Birth: 23rd September 1984

Marital Status: Married

Husband Name: Dr. Shankar Lal J. (Gen.Surgeon)

Language Known: English, Hindi, Marathi, Gujarati, Rajasthani, Kannada

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Research Interests

Organic/medicinal chemistry, structure-based drug design, organic synthesis; development of analytical methods and preformulation studies, herbal preparations, quality assurance, nutraceuticals, pharmaceutical production.

Education

- **Ph.D. in Medicinal Chemistry, 2008-2012**
 - **Institutes:** Birla Institute of Technology and Science, Pilani in collaboration with University of North Carolina, Greensboro, US, University of Toronto and University Health Network, Toronto, Canada.
 - **Supervisor:** **Prof. Lakshmi P. Kotra**, University of Toronto, Canada; Scientist and Principal Investigator at University of Health Network, Toronto, Canada; Director of Centre of Molecular Design and Preformulation.
 - **Doctoral dissertation:**
Title: Design and synthesis of small molecule probes targeting malaria, diabetic neuropathy and thrombocytopenia

Part 1. Design and synthesis of CMP derivatives to investigate the mechanistic details of orotidine-5'-monophosphate decarboxylase (ODCase).

Part 2. Phenoxy propanediol derivatives targeting neuroaxonal growth in diabetic neuropathy.

Part 3. Design and synthesis of novel pyrazole derivatives inhibiting Fcγ receptor mediated phagocytosis in idiopathic immune thrombocytopenia (ITP).
- **Master of Pharmacy, 2006-2008 (CGPA 9.13/10)**
 - **Institute:** Birla Institute of Technology and Science, Pilani
 - **Courses taken during M.Pharmacy**
Compulsory Courses: Dosage Form Design, Advanced Medicinal Chemistry, Advanced pharmacology, Quality Assurance and Regulatory Affairs, Clinical Pharmacy and Therapeutics, Professional Practice-I, Pharmacokinetics and Clinical Pharmacy, Pharmaceutical Biotechnology, Professional Practice-II, Instrumental Method of

analysis.

Elective Courses: Advanced Physical Pharmaceutics, Pharmaceutical Management and Administration.

- *Bachelor of Pharmacy, 2002-2006 (Distinction, 76.71%)*
 - *Institute:* University of Mumbai
- *Intermediate (XIIth), 2000-2002 (Distinction, 80.67%)*
 - *Institute:* Ramniranjan Jhunjhunwala Junior College, Mumbai
- *High School (Xth), 1999-2000 (Distinction, 82.80%)*
 - *Institute:* Hindi High School English Medium, Mumbai

Research Experience

- Guided and supervised post graduate Ayurvedic students for their research projects [Rasashatra and Dravyaguna projects] at Shree Jagadguru Gavisiddheshwar Ayurvedic Medical College, Post Graduate Studies & Research Center, Koppal, Karnataka.
- Established Analytical Laboratory at Shree Jagadguru Gavisiddheshwar Ayurvedic Medical College, Post Graduate Studies & Research Center, Koppal, Karnataka
- Guided three master students at Maliba Pharmacy College
 1. *Patel Vidhi Satishkumar:* Stability Indicating HPTLC method for Estimation of Glibenclamide in its Pharmaceutical dosage forms.
 2. *Chovatiya Shraddha Bhikhalal:* Development and validation of stability indicating HPTLC method for simultaneous estimation of ciprofloxacin hydrochloride and Tinidazole in combined dosage forms.
 3. *Pajwani Shefali Vijaykumar:* Development and characterization of microencapsulation of citral rich essential oil
- Research Scholar at University Health Network, Toronto, Canada, **2009-2012.**
 - Medicinal Chemistry-structure based drug design.
 - Multi step organic syntheses using various methodologies.
 - Microwave assisted coupling reactions.
 - Synthesis of heterocyclic compounds.
 - Synthesis of bioaffinity matrix and optimization of serial affinity chromatography technique to isolate the druggable protein from sensory neuronal cell extract.
 - Development of preclinical formulations for in-vivo screening in ITP mouse model.
 - Determination of physicochemical properties (pKa, solubility, Log P) for new chemical entities (NCEs) on Sirius T3.
- Research Scholar at University of North Carolina, Greensboro, US, **2008-2009.**
 - Design and synthesis of pyrimidine nucleosides and nucleotides as

enzyme inhibitors.

- Research trainee at BITS Pilani, **2007**.
 - Synthesis and screening of salicylamides for antimycobacterial activity.

Teaching Experience

- Principal at D.R.Karigowda College of Pharmacy. Hassan November 2018 till to date.
- Associate Professor at D.R.Karigowda College of Pharmacy, Hassan July 2018
- Assistant Professor at Sri Adichunchanagiri College of Pharmacy, B.G.Nagara, Karanataka **2017-2018**
- Research Co-coordinator and Guest Lecturer at Shree Jagadguru Gavisiddheshwar Ayurvedic Medical College, Post Graduate Studies & Research Center, Koppal, Karnataka **2016-2017**
 - Subjects Taught:
 1. Research Methodology for Post Graduate Scholars and IVth year BAMS students
 2. Pharmacology for IInd year BAMS students
- Assistant Professor at Maliba Pharmacy College, Gujarat **2013-2014**
 - Subject Taught: Theory and Practical - Medicinal Chemistry (2nd and 3rd Year), Organic Chemistry (1st and 2nd year). Pharmaceutical Analytical Practical for Postgraduate students.
- Research Scholar at University Health Network, Toronto, Canada, **2009-2012**.
 - Supervised and taught graduate and undergraduate students in organic chemistry and preformulation labs.

Industrial Experience

- Research trainee at Wockhardt Research Centre, Aurangabad, Maharashtra, **2008** (5 ½ months).
 - Development and optimization of wet milling technology to improve bioavailability of class II drugs.
- Summer trainee at Cipla, Mumbai, **2005**.
 - Completed one month industrial training programme.

Technical Skills

- Instrumentation handling, operation and maintenance
 - NMR-Joel 600MHZ, Bruker 400MHZ
 - ATR-IR (Bruker)
 - Sirius T3 (Used for determination of physicochemical of properties pKa, solubility, Log P)
 - Microfluidizer
 - UV-visible spectrophotometer (Molecular Devices)
 - Biotage® Initiator (Microwave Synthesizer)
 - Biotage® Isolera™ (Flash Purification System)
 - Particle size analyzer (Beckman Coulter)
 - Confocal Raman microscope (Horiba)
 - Ball mill (Nano mill) (Beckman Coulter)
 - HPLC (Waters)
 - LCMS system (Waters)
 - ReactIR™iC10 (Mettler Toledo)
 - Hot-Stage Inverted Microscope (Nikon)

- **Analytical techniques**

- NMR interpretation– ¹H, ¹³C, COSY, HMQC, HMBC, DEPT,NOESY
- Mass spectrometry
- Titrations
- Chromatography-thin layer, Flash Chromatography, HPLC
- Spectrophotometry- UV, IR, Raman
- Gel electrophoresis-SDS-PAGE

Computer Skills

- Office applications-Microsoft PowerPoint, Word, Excel.
- Technical drawing-Photoshop.
- Scientific applications- SciFinder, Reaxys (Beilstein), Chemdraw, COOT, Discovery Studio.

Professional Skills

- Resource Person in the Research Methodology Workshop at SJGAMC **2016**
- Editor of Annual Report 2013-14 of UKA Tarsadia University
- Actively involved in the installation, training and maintenance of equipments for preformulation laboratory at University Health Network, Canada, **2010-2012**.
- Databases: PubMed/MedLine (US National Library of Medicine); SciFinder Scholar, Reaxys.
- Strong communication, presentation and scientific writing (research papers, reports, SOPs) skills.
- Worked with multidisciplinary research group comprising of people of diverse origins and cultures. Can work independently and in team settings.
- Hardworking, quick-learner and gets job done on time.

Academic Achievements

- Selected as exchange student from BITS Pilani for doctoral dissertation, **2008**.
- College topper for all the four years of B.Pharmacy.
- Bagged first prize in “State level” pharmacy quiz-**2006** held at SVB college of pharmacy, Mumbai.
- Awarded as “Best student” during Pharmacy, **2006**.
- Merit Scholarships awarded by “Ratan Tata Trust” for all the four years of B.Pharmacy.

Conferences and Presentations

1. Attended conference, “**Generics and Biologics 2014**”, held at Ramada Powai Hotel & Convention Centre, Mumbai. Select Biosciences India Private Limited
2. Purohit, M. K.; Poduch, E.; Wei, L.; Crandall, I.; Kain, C. K.; Pai, E.; Kotra, L. P. N^3 - and N^4 -CMP derivatives as orotidine -5- monophosphate decarboxylase inhibitors. Canadian Society for Pharmaceutical Sciences (CSPS), **2012**, Toronto, Canada. (Poster Presentation).
3. Purohit, M. K.; Calcutt, A. N.; Fernyhough, P.; Kotra, L. P. Effect of chirality on neurite outgrowth of sensory neurons using phenoxy propanediol derivatives. Canadian Society for Pharmaceutical Sciences (CSPS), **2012**, Toronto, Canada. (Poster Presentation).
4. Purohit, M. K.; Poduch, P.; Pai, E.; Kotra, L. P. Unconventional derivatives of CMP as inhibitors of Orotidine-5'-monophosphate decarboxylase-an antimalarial target. Indian Pharmaceutical Congress (IPC), **2010**, Manipal, India. (Best Paper Award, oral presentation).
5. Purohit, M.K.; Menon, N.; Senthil; Yogeswari; D. Sriram. A facile microwave assisted synthesis of novel 3,5-dinitrosalicylamides and their antimycobacterial activity. Indian Pharmaceutical Congress (IPC), **2007**, Varanasi, India.
6. Participated in workshop of “Targeted & Controlled Drug Delivery,” organized by VMHPS college of Pharmacy, **2006**, Mumbai.

Extra-curricular Activities /Hobbies

- Vice-president of Indian association “YUVA”, at University of North Carolina, Greensboro, US, **2008-2009**.
- Bagged first position in seminar and debate during B.Pharmacy.
- Hobbies-reading, writing and travelling.

Publications

1. Bello, A. M.; **Purohit, M. K.**; Cui, Y. J. T.; Stead, B. S.; Kotra, L. P. Immunological targets in inflammation from the small molecule perspective. *Anti-Inflamm. Anti-Allergy Agents Med. Chem.* **2011**, 10, 121-131.
2. **Purohit, M. K.**; Poduch, E.; Wei, L. W.; Crandall, I. E.; To, T.; Kain, K. C.; Pai, E. F.; Kotra, L. P. Novel cytidine-based orotidine-5'-monophosphate decarboxylase inhibitors with an unusual twist. *J. Med. Chem.* **2012**, 55 (22): 9988-97
3. Hadimani, M.; **Purohit, M. K.**; Vanampally, C.; Pleog, R.; Abrallo, V.; Morrow, D.; Frizzi, K.; Calcutt, N.; Fernyhough, P.; Kotra, L. P. Guaifenesin derivatives promote neurite outgrowth of sensory neurons and protect diabetic mice from neuropathy. *J. Med. Chem.* **2013**, 56(12): 5071-8.
4. **Purohit, M. K.**; Scovell, I.; Neschadim, A.; Katsman, Y.; Branch, R. D.; Kotra, L. P. Disulfide Linked Pyrazole Derivatives Inhibit phagocytosis of opsonized blood cells phagocytosis. *Bioorg. Med. Chem. Lett.* **2013**, 23(8):2323-7
5. **Purohit, M. K.**; Chakka, K. S.; Scovell, I.; Neschadim, A.; Katsman, Y.; Branch, R. D.; Kotra, L. P. Structure-activity relationships of pyrazole derivatives as potential therapeutics for immune thrombocytopenias. *Bioorg. Med. Chem.* **2014**, 22(9): 2739-52
6. Bello, A. M.; Wei, L.; Lewis, M. M.; Beata Majchrzak-Kita, **Purohit, M. K.**; Fish, N. L.; Kotra, L. P. Small molecule mimetics of an interferon-alpha-receptor interacting domain. *Bioorg. Med. Chem.* **2014**, 22(3): 978-85.

References

Dr. Lakhmi P. Kotra

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